



Peripherally Inserted Central Catheter – Care of Procedure

1. Scope

Site	Operational Area	Applicable to
Armada Kalamunda Group	All clinical services	Nursing and Midwifery

2. Purpose

This document provides guidance for clinicians to provide optimal care for patients having a peripherally inserted central catheter.

Insertion of PICC lines will only be undertaken by:

- Clinical Nurse Specialist Vascular Access or a nurse suitably trained by the Clinical Nurse Specialist Vascular Access
- Medical staff – registrar level or above with appropriate training

Care and removal of PICC lines will only be undertaken by persons with the appropriate training in the procedure, or under the direct supervision of an appropriately trained person. For nurses/midwives this means successful completion of the PICC self-directed learning package.

All PICCs are to be reviewed daily and removed when no longer clinically indicated

- The veins of the forearm, upper arm and subclavian are of critical importance for the creation of arterio-venous fistula for haemodialysis and are therefore not routinely used for venous access.
- The Renal Physician/Nephrologist must be consulted for patients with chronic renal impairment.
- Staff are to avoid taking blood pressure on the arm with the PICC line due to risk of occlusion.

3. Procedure

3.1 General management

- Check that the inserting practitioner has documented the tip is in a suitable position and the line is ready for use.
- Assess insertion site and dressing integrity each shift and record PIVAS score.
- PICC dressing to be changed seven days post insertion, unless blood soaked beforehand.
- Weekly dressing changes are required thereafter or PRN if wet, soiled or blood soaked. . All components of the PICC line are changed with every dressing change.

- Utilise a sutureless fixation device such as Statlock or 3M Securement Device to secure the line at the insertion point.
- Cover the PICC line with a transparent dressing to protect the site from extrinsic contamination and allow observation.
- Use a dressing with a chlorhexidine gel pad or place a chlorhexidine impregnated sponge (such as Biopatch with blue side up) on the insertion point. It has a residual anti-microbial benefit for 7 days.
- Hand Hygiene: careful handling, securing of lines and aseptic technique and the use of alcohol impregnated swabs to prime the line before use will reduce infection, maintain patency and promote longevity.
- Only use in-situ clamps when changing positive pressure valve (PPV) bung.
- Flush the line pre and post **ALL** infusions/injections using 10ml Sodium Chloride 0.9% in a 10ml syringe in a push-pause method to maintain patency and positive pressure within the catheter.
- Flush all the lumens of the catheter if more than one, including the lumens that have not been used.
- Flush PICC line using push-pause method immediately post any blood sampling, with 20ml sodium chloride, using the push pause method to increase the turbulence in the line and remove any build-up of particulate or blood.
- All PICC lines should be fitted with a Positive Pressure Valve (such as MaxPlus clear) bung to prevent line blockage.
- Blood products can be transfused via PICC but must be flushed with 20mls of Sodium Chloride 0.9% post infusion.
- If PICC blocked or leaking do not remove. Contact the Vascular Access Service on 042 624 7182 immediately.
- Cover PICC dressing with waterproof cover when showering and change dressing if wet or soiled.
- Send PICC tip on removal for microscopy, culture and sensitivity **only** if PICC infection is suspected. **Note** peripheral blood cultures **must** be taken and sent at the same time.
- Do a swab for MC&S on the insertion site if infection is suspected before the PICC is removed.

Do not use syringes smaller than 10ml as this may cause line rupture.

Do not perform blood pressure measurement with inflating cuff on arm with PICC insitu.

3.2 PICC Daily Review and Dressings

3.2.1 Dressing Changes:

- Frequency is determined by dressing type and condition.
- Every 7 days for transparent, semipermeable dressing.
- Every 2 days for sterile gauze dressing.
- Change if dressing integrity is compromised or if moisture or blood is built up.

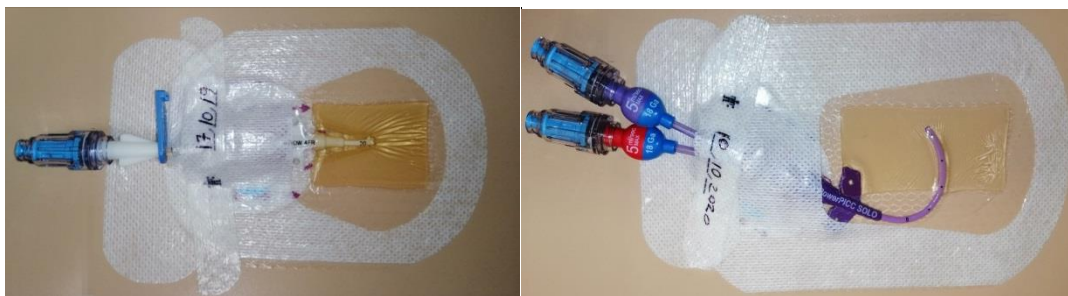
- The catheter insertion site should be visually inspected or palpated through the intact dressing to determine tenderness during daily review. The process of dressing changes, when indicated, provides the opportunity closely review and cleanse the insertion site.
- Suture less securement device if insitu requires changing every 7 days.
- Positive Pressure Valves should be changed every 7 days.
- Cover VAD dressing with waterproof cover when showering.

3.2.2 Equipment Required

- 10ml syringe
- Sterile transparent semi-permeable IV Dressing with Chlorhexidine gel such as 3M Tegaderm CHG **OR** Chlorhexidine gluconate impregnated foam disk and plain semi-permeable IV Dressing.
- Alcohol swabs
- Ampoule sodium chloride 0.9%
- Antimicrobial swab sticks such (e.g.:BD Persist Plus swab sticks or 3M SoluPrep Swab) OR antimicrobial Chlorhexidine Ethanol surgical prep solution
- Skin barrier film swab stick (e.g.: 3M Cavilon No sting barrier film).
- PPE: sterile & non-sterile gloves
- SmartSite Plus positive pressure valve (e.g.: Maxplus clear)
- Sterile dressing pack
- Suture less securement device (e.g.: 3M Securement Device- comes with 3M Tegaderm CHG dressing).Tubular bandage to cover limb

3.2.3 Procedure

- Perform hand hygiene, clean work surface and assemble equipment.
- Position patient to expose PICC insertion site.
- Perform hand hygiene, don gloves.
- Remove transparent dressing, peeling from distal to proximal end to avoid dislodging PICC.
- Using alcohol swab, remove suture less securement device. Ensure PICC is stable and **do not leave the patient unattended.**
- Discard dressing and gloves. Perform hand hygiene, don sterile gloves.
- Place sterile towel under PICC.
- Clean PICC insertion site with antiseptic solution until all visible debris is removed. **Allow to air dry.**
- Clean entire area that will be covered by dressing.
- Apply skin barrier film (e.g.: 3M Cavilon No sting barrier film).
- Replace suture less securement device, as per product information.



- Apply transparent IV Dressing with Chlorhexidine gel. Ensure gel pad covers insertion site. Write date of dressing performed.
- Prime positive pressure valve with Sodium Chloride 0.9%, remove used valve and attach primed valve to line and flush with Sodium Chloride 0.9% in a 10ml syringe using the PUSH/PAUSE technique.
- Apply tubular bandage over dressing if appropriate.
- Record date of dressing change, length of exposed catheter and next dressing due on PICC Care Plan.
- Dispose of waste. Remove gloves and perform hand hygiene

3.2.4 PICC Dressing Post-Procedure

- Ensure infusion lines are appropriately secured.
- Document details of procedure in the medical record

3.3 PICC Flushing

Frequency: Regular flushing regimes for PICC to be determined based on considerations regarding the individual device and attachments being used, and the clinical requirements and situation of the patient.

Method: Administer flush 1 mL at a time, using a pulsating action (brisk push-pause) to create turbulence within the catheter, and end with applying positive pressure within the lumen.

Do not forcibly administer flush. Consult with Vascular Access Service for advice.

3.3.1 Indications:

- To assess and maintain PICC patency prior to infusions.
- Following infusions to prevent mixing of incompatible medications or solution

3.4 Changing Positive Pressure Valve (PPV)

3.4.1 Indications

- Replace PPV every seven days with dressing changes, if contaminated, or leaking.
- Synchronise PPV change with flushing interval, where appropriate.

3.4.2 Equipment Required

- 10 ml Luer Lock Syringes

- Closed end syringe cap (all prepared syringes are to be capped during transport to the patient).
- Kidney dish
- Non-sterile gloves
- Positive pressure valve
- Select the amount, and type of flushing solution suitable to the situation.
- Sterile solution for injection:
 - Sodium Chloride 0.9%
 - 5% dextrose followed by Sodium Chloride 0.9% if medication incompatible with Sodium Chloride 0.9%

3.4.3 Procedure

- Perform hand hygiene and assemble equipment. Draw up syringes. Prime positive pressure valve (PPV).
- Perform hand hygiene.
- Working one lumen at a time.
- Clamp PICC line if it is non-valved (e.g.: Arrow PICC lines).
- Remove and discard existing PPV using aseptic no touch technique, taking care not to contaminate the exposed PICC hub and replace swiftly with new primed PPV.
- Insert flush syringe into PPV, flush the lumen using a pulsatile method (push-pause). Repeat for each lumen.
- Discard equipment.
- Perform hand hygiene.
- Document details of procedure.

3.5 PICC Blood Sampling (Including Blood Cultures)

3.5.1 Indications:

- For patients where no other alternative exists.
- Following placement of a new device.

3.5.2 Indications for Blood Cultures

To determine suspected central-line blood stream infection, undertake:

- One sample collected from existing device and,
- One sample collected from separate peripheral venepuncture

3.5.3 PICC Blood Sampling Advice

- Utilise the largest lumen available.
- If multiple lumen catheter insitu, dedicate one lumen to purpose.
- Undertake blood sampling from lumen with no continuous infusion in progress.
- Flush line prior to blood sampling, as per site requirements, and always follow blood sampling policy.
- Flush all lumens post blood sampling, including those not used.

3.5.4 Equipment

- Blood Collection device(one of the following):
- Vacutainer safe adapter
- 10ml syringe + vacutainer blood transfer device.
- 10 ml luer-lock syringe
- 20ml luer-lock syringe
- Antiseptic solution swabs
- Kidney dish
- Non-sterile gloves
- Positive pressure valve
- Sodium Chloride (0.9%) for injection 20ml minimum (for flushing)
- Specimen request form
- Vacuumed Aerobic and anaerobic blood culture specimen bottles
- PICC catheter access hub
- For blood cultures:
 - Vacutainer safe adapter
 - 20ml luer-lock syringe
 - Alcohol Swabs
 - Blood culture Bottles

3.5.5 Procedure

- Confirm correct patient details with pathology request form.
- Perform hand hygiene, clean work surface and assemble equipment.
- Cease infusion if in progress and if clinically appropriate.
- Perform hand hygiene. Don gloves.
- Clean PPV with antiseptic swabs. **Allow to air dry.**
- Flush lumen with 10 mL Sodium Chloride 0.9%, attach 10 mL syringe into PPV and withdraw 5 mL blood.
- Discard syringe containing blood.
- Clean PPV with antiseptic swabs. Allow to air dry.
- Then withdraw the required amount of blood, by either attaching:
 - Assembled vacutainer holder + direct draw adapter, and inserting blood specimen tubes.
 - 10 mL (minimum) syringe and 20 mL syringe for blood culture.
- Remove blood device from PPV and place samples in kidney dish.
- Flush all lumens with 20 mL sodium chloride 0.9% using a pulsatile method (push-pause) until lumen and PPV are clear.
- Clean PPV with antiseptic swabs. Allow to air dry.
- For blood collected via syringe:
 - Blood cultures: clean the top of blood culture bottles with alcohol swabs. Attach vacutainer safe adapter to syringe, place 10 mL of blood into bottle. Repeat with new vacutainer safe adapter for second bottle.

- Specimen tubes: attach vacutainer safe adapter place appropriate amount of blood in each tube.
- Ensure VAD is secure.
- Discard equipment. Perform hand hygiene.

3.5.6 Post Procedure

- Label samples with patient identification sticker, date/time sample taken. For blood cultures, label sample taken from device. Complete request form and send samples to laboratory.
- For blood cultures, label sample taken from device. Complete request form and send samples to laboratory
- Document details of procedure.

3.6 PICC Removal Equipment

- Alcohol Swabs
- Dressing Pack
- Dressing Solvent (if available)
- PPE goggles/visor, sterile and non-sterile gloves
- Small occlusive dressing
- Skin antiseptic solution

3.7 PICC Removal Procedure

- Perform hand hygiene. Prepare equipment.
- Position patient in supine, as tolerated.
- PICC: support and extend arm 45-90 degrees.
- Perform hand hygiene. Don PPE.
- Blood cultures: Undertake collection here; refer to PICC & Blood Sampling
- Using dressing solvent if available, remove and discard dressing. Remove securement device using alcohol swabs and discard. Do not leave patient.
- Perform hand hygiene (apply alcohol gel and allow to air dry). Don sterile gloves.
- Create sterile field using aseptic non-touch technique: underneath the arm.
- Clean device insertion site. Allow to air dry.
- Place gauze over device insertion site in preparation for removal.
- Request patient to hold breath and on expiration withdraw catheter and place onto sterile field:
- Withdraw with slow intermittent traction, without applying tension.
- If resistance is felt, stop removal, cover site with sterile dressing and apply dry warm pack to upper arm for 15 to 20 minutes. If resistance still felt after this time consult medical officer or Vascular Access nurse
- On removal apply gentle pressure with gauze on the site until haemostasis is achieved.
- Apply sterile occlusive dressing, and remain intact for a minimum of 24 hours.
- Observe device integrity

3.8 Documentation

For all PICC procedures ensure to document:

- Date, time and details specific to the procedure undertaken
- Actual type, size and distinctive features of the device inserted or accessed
- Details of aseptic technique, e.g. emergency insertions
- If any modifications have been made to the device, e.g. shortened for insertion
- Insertion site identifiers, i.e. anatomical location, and/or marked drawings
- Any attempted or incomplete procedures, e.g. unsuccessful insertions
- Outcome of insertion site assessment
- Details of the device patency and condition
- Rationale regarding the need for device commencement, continuation or removal
- Complete any required site specific audit tool, and include within the patient health record, e.g. PICC insertion audit sticker care.

4. Legislative context

The following documents are required to give effect to this procedure:

- Health Department of Western Australia (HDWA) [Consent to Treatment Policy OD 0175/22](#)
- HDWA [Clinical Handover Policy MP 0095](#)
- East Metropolitan Health Service (EMHS) [Patient Identification and Procedure Matching EMHS:](#)

5. Supporting information

The following documents support the implementation of this procedure:

- [Hand Hygiene Policy AKG-POL-0187-17](#)
- [Aseptic Technique Procedure AKG-PRO-045-14](#)
- [Blood and Blood Product Management Guideline AKG-GUI-0060-11](#)

6. Compliance, monitoring and evaluation

Compliance with this policy will be monitored via:

- Monitoring of reported clinical incident data via Datix CIMS.
- Observational audit of aseptic technique at the point of care following invasive procedure assessed as high risk:
 - CVC/PICC insertion, access and management

Department specific findings are to be reported at the relevant departmental meeting for review and identification of any opportunities for improvement.

Organisation-wide data trends and risks associated with PICC are to be reported to, and monitored by, the Clinical Communications Committee.

7. Relevant standards

National Safety and Quality Health Service (NSQHS) Standards (second edition):

- Standard 3 – Action 3.8 Hand hygiene
- Standard 3 – Action 3.9 Aseptic technique
- Standard 3 – Action 3.10 Invasive medical devices
- Standard 7 – Action 7.6 Prescribing and administering blood and blood products

8. References

1. Corley, A., Ullman, A. J., Mihala, G., Ray-Barruel, G., Alexandrou, E., & Rickard, C. M. (2019). Peripheral intravenous catheter dressing and securement practice is associated with site complications and suboptimal dressing integrity: A secondary analysis of 40,637 catheters. *International Journal of Nursing Studies*, 100, 103409. <https://doi.org/10.1016/j.ijnurstu.2019.103409>
2. Gao, Y., Liu, Y., Zhang, H., Fang, F., & Song, L. (2018). The safety and accuracy of ECG-guided PICC tip position verification applied in patients with atrial fibrillation. *Therapeutics and Clinical Risk Management*, 14(1), 1075-1081. <https://doi.org/10.2147/tcrm.s156468>
3. Gorski, L. A. (2017). The 2016 infusion therapy standards of practice. *Home Healthcare Now*, 35(1), 10-18. <https://doi.org/10.1097/nhh.0000000000000481>
4. Günther, S. C., Schwebel, C., Hamidfar-Roy, R., Bonadona, A., Lugosi, M., Ara-Somohano, C., Minet, C., Potton, L., Cartier, J., Vésin, A., Chautemps, M., Styfalova, L., Ruckly, S., & Timsit, J. (2016). Complications of intravascular catheters in ICU: Definitions, incidence and severity. A randomized controlled trial comparing usual transparent dressings versus new-generation dressings (the advanced study). *Intensive Care Medicine*, 42(11), 1753-1765. <https://doi.org/10.1007/s00134-016-4582-2>
5. Jackson, A. (1998). Infection control – a battle in vein: infusion phlebitis. *Nursing Times*, 94(4), 68-71.
6. Li, A., Jiao, J., Zhang, Y., Tian, L., Miao, J., Hao, X., Sun, Z., & Sun, Q. (2018). A randomized controlled study of bedside electrocardiograph-guided tip location technique & the traditional chest radiography tip location technique for peripherally inserted central venous catheter in cancer patients. *Indian Journal of Medical Research*, 147(5), 477. https://doi.org/10.4103/ijmr.ijmr_1120_16
7. Ray-Barruel, G., Polit, D. F., Murfield, J. E., & Rickard, C. M. (2014). Infusion phlebitis assessment measures: A systematic review. *Journal of Evaluation in Clinical Practice*, 20(2), 191-202. <https://doi.org/10.1111/jep.12107>
8. Walker, G., Alexandrou, E., Rickard, C. M., Chan, R. J., & Webster, J. (2015). Effectiveness of electrocardiographic guidance in CVAD tip placement. *British Journal of Nursing*, 24(14), 4-12. <https://doi.org/10.12968/bjon.2015.24.sup14.s4>

9. Glossary

The following definitions are relevant to this policy.

Term	Definition
PICC Line	A PICC is a radiopaque central catheter inserted via a vein in the upper arm with the tip residing in the Superior Vena Cava.

10. Policy owner (relating to these procedures)

Enquiries relating to this procedure may be directed to:

Title: Coordinator Nursing
 Department: Hospital Wide
 Email: shane.hastings@health.wa.gov.au

11. Review

This procedure will be reviewed and evaluated as required to ensure relevance and currency. At a minimum it will be reviewed within one (1) year after first issue and at least every three (3) years thereafter.

Version	Effective from	Effective to	Amendment(s)
V3	05/07/2023	05/07/2026	None

12. Authorisation

This procedure has been authorised and issued by the Director Clinical Services.

Authorised by	Director of Clinical Services
Authorisation date	29/06/2023
Published date	05/07/2023